Appendix. Mathematics, deep learning, PyTorch

This appendix offers a rapid review of deep learning, the relevant mathematics we use in this book, and how to implement deep learning models in PyTorch. We'll cover these topics by demonstrating how to implement a deep learning model in PyTorch to classify images of handwritten digits from the famous MNIST dataset.

Deep learning algorithms, which are also called *artificial neural networks*, are relatively simple mathematical functions and mostly just require an understanding of vectors and matrices. Training a neural network, however, requires an understanding of the basics of calculus, namely the derivative. The fundamentals of applied deep learning therefore require only knowing how to multiply vectors and matrices and take the derivative of multivariable functions, which we'll review here. *Theoretical machine learning* refers to the field that rigorously studies the properties and behavior of machine learning algorithms and yields new approaches and algorithms. Theoretical machine learning involves advanced graduate-level mathematics that covers a wide variety of mathematical disciplines that are outside the scope of this book. In this book we only utilize informal mathematics in order to achieve our practical aims, not rigorous proof-based mathematics.